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1. Training MSE = 0.85 (very low error)

Testing MSE = 5.20 (high error)

The large difference between traning and testing errors indicates that neural networks is overfiting. It has learned the training data verywell but does not generalize effectively to new, unseen data

~~40~~ 45/50

90%

~~80%~~

Possible Reasons

1. over fitting - The Model memorizes the training data instead of learning general patterns.

2. Model is too complex: To many hidden layers or neurons for available data.

3. insufficient training data: A small dataset makes it harder to generalize

4. Noisy or inconsistent data: Errors or outliers in dataset affect performance.

5. Lack of regularization: The model is not constrained leading to excessive fitting.

Corrective Measures

Use more training data to improve generalization

Apply regularization (L1/L2 regularization or dropout)

- Reduce model complexity by using fewer layers or neurons
- use early stopping to stop training before overfitting occurs.
- perform cross-validation to evaluate the model more reliably.
- clean and preprocess the data by removing noise and handling outliers.

conclusion:

since training MSE (0·85) is much lower than testing MSE (5·20), the neural neural network is out overfitting improving the dataset, simplifing the model, and applying regularization techniques can reduce the testing error and improve prediction accuracy.